

YU XIAN, China

WMO: 535930

Lat: 39.8359N

Lon: 114.5723E

Elev: 2986

StdP: 13.18

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.2	(c) -0.3	(d) -18.4	(e) 2.2	(f) 6.1	(g) -15.3	(h) 2.7	(i) 8.7	(j) 16.7	(k) 22.9	(l) 14.4	(m) 20.8	(n) 1.9	(o) 250	(p) 0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	90.1	64.9	87.5	64.7	85.1	64.3	71.8	82.4	70.3	80.5	68.9	78.7	6.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.6	117.5	77.0	67.1	111.6	75.9	65.5	105.4	75.0	37.7	82.3	36.4	80.6	35.0	78.7	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	14.2	12.0	-10.4	95.2	3.8	3.5	-13.1	97.7	-15.3	99.7	-17.4	101.7	-20.1	104.2
			-11.0	74.2	3.7	1.3	-13.7	75.1	-15.8	75.9	-17.9	76.6	-20.5	77.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	15.3	24.0	37.9	51.9	63.0	71.2	74.8	71.7	62.3	49.5	33.4	19.5
	DBStd	21.71	7.54	8.42	8.65	8.50	6.80	5.12	4.43	4.52	6.02	7.35	8.49	7.68
	HDD50	3813	1077	727	385	77	3	0	0	0	1	98	499	946
	HDD65	7021	1542	1147	841	399	119	10	1	5	117	481	948	1411
	CDD50	3082	0	0	9	134	407	636	769	672	371	83	1	0
	CDD65	815	0	0	0	6	58	197	305	212	36	1	0	0
	CDH74	6948	0	0	1	129	697	1737	2530	1495	342	17	0	0
	CDH80	2164	0	0	0	29	194	580	900	390	70	1	0	0
(14) Wind	WSAvg	4.3	3.3	4.0	5.2	5.8	5.8	4.9	4.1	3.7	3.7	3.8	3.6	3.5
(15) Precipitation	PrecAvg	16.1	0.1	0.2	0.4	0.8	1.4	2.4	4.3	3.5	1.8	0.9	0.3	0.1
	PrecMax	24.3	0.4	0.7	1.5	3.7	3.0	5.0	7.8	6.9	3.5	2.3	0.8	0.4
	PrecMin	8.8	0.0	0.0	0.0	0.1	0.1	0.6	1.9	0.7	0.2	0.0	0.0	0.0
	PrecStd	3.9	0.1	0.2	0.3	0.7	0.7	1.2	1.5	1.6	0.9	0.6	0.2	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	53.4	69.6	84.2	89.5	92.7	94.4	90.5	86.7	76.5	61.0	43.2
		MCWB	30.4	37.6	46.7	53.9	57.8	62.6	67.4	68.2	62.2	56.0	44.2	32.8
	2%	DB	36.0	46.8	62.6	77.3	85.2	89.2	90.7	86.7	81.4	71.1	56.1	38.6
		MCWB	27.2	34.0	43.4	52.0	57.1	62.1	67.1	67.5	60.6	52.8	41.4	29.3
	5%	DB	32.4	42.9	57.7	72.5	81.4	86.3	87.8	84.2	77.7	67.0	51.4	35.1
		MCWB	24.9	31.9	40.9	49.7	55.5	61.4	67.3	66.9	59.6	50.7	39.4	26.9
	10%	DB	28.7	38.9	53.2	68.1	77.7	83.4	85.2	81.8	74.5	63.2	47.1	32.1
		MCWB	22.3	29.7	38.6	48.1	54.4	60.9	67.3	66.2	58.3	49.2	37.5	25.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.0	39.2	47.9	57.9	62.6	68.7	74.2	73.1	67.0	58.4	46.1	33.3
		MCDB	39.4	49.6	65.2	73.0	78.6	82.5	86.6	83.5	77.9	71.5	54.2	43.0
	2%	WB	27.7	35.3	44.7	54.8	60.6	66.9	72.3	71.6	64.5	55.1	43.3	29.7
		MCDB	35.3	45.3	60.1	71.1	75.8	80.0	83.4	81.6	74.6	65.6	53.5	37.5
	5%	WB	25.2	32.5	42.0	52.1	58.8	65.5	71.0	70.1	62.3	53.0	40.7	27.5
		MCDB	32.1	41.6	56.2	67.9	74.3	78.1	81.7	79.3	72.9	63.4	50.0	34.4
	10%	WB	22.5	30.1	39.4	49.8	57.0	64.0	69.6	68.6	60.5	50.9	38.0	25.4
		MCDB	28.4	38.4	52.2	65.4	72.4	76.7	80.1	77.4	70.9	60.7	46.8	31.9
(35) Mean Daily Temperature Range	5% DB	MDBR	21.1	21.9	22.0	23.3	22.6	20.8	17.9	18.2	20.5	20.9	20.0	19.3
		MCDBR	24.6	27.1	29.0	30.7	29.1	25.8	22.5	21.5	25.9	26.7	26.6	23.3
		MCWBR	17.7	17.3	15.0	12.5	10.1	7.6	6.4	6.9	8.8	11.2	14.7	15.9
	5% WB	MCDBR	23.7	25.3	27.0	25.6	22.4	20.0	17.7	16.8	19.7	22.0	24.2	22.1
		MCWBR	17.2	16.2	14.3	11.1	8.8	7.0	6.6	6.5	8.1	10.1	13.9	15.3
(40) Clear-Sky Solar Irradiance	taub		0.357	0.445	0.475	0.544	0.595	0.708	0.690	0.676	0.584	0.509	0.392	0.358
	taud		2.077	1.873	1.837	1.751	1.667	1.514	1.590	1.594	1.716	1.818	2.051	2.119
	Ebn at Noon		244	233	243	236	227	202	204	201	210	210	228	233
	Edh at Noon		36	52	60	70	77	90	83	81	67	54	37	33
(44) All-Sky Solar Radiation	RadAvg		826	1076	1483	1753	1954	1761	1645	1578	1372	1117	846	731
	RadStd		52	70	120	112	145	138	93	106	89	89	100	58

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page